This is a test document about DNA and genetics for the AI learning platform.

DNA is the hereditary material in humans and almost all other organisms. DNA stands for deoxyribonucleic acid. Most DNA is located in the cell nucleus, but a small amount can also be found in the mitochondria.

DNA is made up of four chemical bases: adenine (A), guanine (G), cytosine (C), and thymine (T). The order of these bases determines the genetic instructions for building and maintaining an organism.

The DNA double helix consists of two strands that twist around each other. The two DNA strands are antiparallel, meaning they run in opposite directions.

DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules. Replication is an essential process because every time a cell divides, the two new daughter cells must contain the same genetic information as the parent cell.

Genes are segments of DNA that contain the instructions for making proteins. Each gene provides the code for a specific protein or a set of proteins.